

1) Recap on P.T.

Events $\subseteq$ Outcome $\subseteq \Omega$ (PT space)

$$P[\emptyset] \leq P[A] \leq P[\Omega] = 1$$

Probability Union

$$P[A \cup B] = P[A] + P[B] - P[A \cap B]$$

Disjoint:

$$A_i \cap A_j = \emptyset \quad \forall i \neq j : \bigcup_i A_i = \Omega$$

Law of total probability:

$$P[B] = \sum_i P[A_i \cap B]$$

Independence:

$$P[A \cap B] = P[A]P[B] \implies A, B \text{ independent}$$

Cumulative Distribution Function (CDF):

$$F(x) = P[X \leq x] = \int_{-\infty}^x p_X(x) dx$$

- $F(x) \in [0, 1]$
- Non decreasing
- *Right* continuous

Expectation:

$$\mathbb{E}[g(X)] = \sum_i g(x_i) P[x_i]$$

spooky scary stuff (this notation means sum in discrete case and int in continuous)

$$\mathbb{E}[g(x)] = \int_{-\infty}^{\infty} g(x) dF(x)$$

Moments:

$$\mathbb{E}[X^m] = \sum_i x_i^m P[x_i]$$

- $m = 1$ is the mean
 - $m = 2$ is the power
- Central moment is adjusted for mean, with $m = 2$ it is the variance

$$\mathbb{E}[(X - \mu)^2] = \text{var}(X)$$

Joint distribution and marginals:

$$F(x; y) = P[X \leq x; Y \leq y]$$

$$f_X(x) = \int f(x, y) dy \quad f_Y(y) = \int f(x, y) dx$$

if

$$F(x; y) = F[x]F[y] \implies iid$$

if

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] \implies \text{uncorrelated}$$

equivalence between iid and uncorrelated holds for gaussian rvs

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

Conditional Probability:

$$P[A|B] = \frac{P[A \cap B]}{P[B]} \text{ for } P[B] \neq 0$$

$$P[A \cap B] = P[A|B]P[B]$$

$$P[A] = \sum_i P[A \cap B_i] = \sum_i P[A|B_i]P[B_i]$$

$$\mathbb{E}[Y] = \sum \mathbb{E}[Y|x_i]P[x_i]$$

Fourier Tx Characteristic Function:

$$\phi(t) = \int e^{it\lambda} dF(\lambda) = \mathbb{E}[e^{itx}]$$

$$\frac{d\phi}{dt} = \mathbb{E}[ixe^{itx}] \stackrel{t=0}{=} it[x]$$

$$\frac{d^2\phi}{dt^2} = \mathbb{E}[(ix)^2 e^{itx}] \stackrel{t=0}{=} i^2 t[x^2]$$

Generating functions:

$$g(s) = \sum_{k=0}^{\infty} p_k s^k = \mathbb{E}[s^x] \stackrel{s=e^{it}}{=} \mathbb{E}[(e^{it})^x]$$

$$g'(s) = \mathbb{E}[xs^{x-1}] \stackrel{s=1}{=} \mathbb{E}[x]$$

$$g''(s) = \mathbb{E}[x(x-1)s^{x-2}] \stackrel{s=1}{=} \mathbb{E}[x^2] - \mathbb{E}[x]$$

then

$$\mathbb{E}[x^2] = g''(1) + g'(1)$$

Bernoulli:

$$x \in \{0, 1\}$$

$$p = P[x = 1] = 1 - P[x = 0]$$

$$\mathbb{E}[x] = p \quad \mathbb{E}[x^2] = p \quad \text{var}(x) = p(p-1)$$

Binomial:

sum of N iid Bernoulli rv

$$y = \sum_{i=1}^n x_i \quad x_i \sim \text{bern}(p)$$

$$\mathbb{E}[y] = n\mathbb{E}[x] = np \quad \text{var}(y) = n\text{var}(x) = np(1-p)$$

with $n \rightarrow \infty$ binomial becomes poisson with $\lambda = np$

Poisson:

$$P[x = k] = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k \geq 0$$

$$\mathbb{E}[x] = \sum_{k=0}^{\infty} k e^{-\lambda} \frac{\lambda^k}{k!} = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} e^{\lambda} = \lambda \quad \text{var}(x) = \lambda$$

Geometric RV:

how many failures before the first success (0,0,0,1 will be 3 not 4) Z

second definition: how many attempts until first success (definition of before +1) $Z + 1$

$$P[Z = k] = (1 - p)^k p$$

$$\mathbb{E}[Z] = \sum_{k=0}^{\infty} k P[Z = k] = \sum_{k=0}^{\infty} P[k > Z] = \frac{1-p}{p} \quad \text{var}(Z) = \frac{1-p}{p^2}$$

Exponential:

$$f_T(t) = \begin{cases} \lambda e^{-\lambda t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$\mathbb{E}[T] = \frac{1}{\lambda} \quad \text{var}(T) = \frac{1}{\lambda^2}$$

Memoryless property:

$$P[T - t > x | T > t] = \frac{P[T - t > x, T > t]}{P[T > t]} = \frac{P[T - t > x]}{P[T > t]} = \frac{e^{-\lambda(t+x)}}{e^{-\lambda t}} = e^{-\lambda x}$$

since $T - t > x$ implies $T > t + x$ and thus $T > t$

Gamma Distribution:

$$\Gamma(\alpha) = (\alpha - 1)!$$

$$f(x) = \frac{\lambda}{\Gamma(\alpha)} (\lambda x)^{\alpha-1} e^{-\lambda x}$$